

Robust Sameness Scoring Using a Shift-Tolerant Correlation Coefficient

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Abstract. Testing whether two two-dimensional ^1H -X HSQC spectra are the *same* — a sameness test that underlies molecular fingerprinting, higher-order-structure comparability of protein biotherapeutics, ligand screening and chemical-shift-perturbation mapping — reduces to a single quantitative comparison, yet the established similarity measures each fail in one of two opposite ways: histogram-binning scores are brittle to the small peak drifts a titration induces, whereas tree- and nearest-neighbour scores saturate because in any crowded window every peak finds a coincidental near neighbour. Here we introduce the Shift-Tolerant Correlation Coefficient (STCC), which renders both spectra onto a single shared grid, blurs each by the physical NMR linewidth combined with the expected shift drift, and scores the pair by the mean-centred, zero-lag normalized cross-correlation of the two images. The Gaussian render supplies graded, physically-tunable shift tolerance while mean-centring actively penalizes intensity that is not co-located. On two deliberately opposite benchmarks — a dense protein ^1H - ^{15}N amide fingerprint and a sparse small-molecule ^1H - ^{13}C set — STCC gives the widest same/different separation among the primary reference methods (0.75 dense, 0.74 sparse on a $[0, 1]$ scale), far outscoring the saturating tree and nearest-neighbour measures. Its dense-fingerprint advantage is large and, on an expanded test, keeps both of two decoy proteins below every same-protein titration point; removing the mean-centring step collapses the original dense-protein separation from 0.75 to 0.59. An experimental, post hoc local-contrast extension raises separation from 0.57 to 0.67 on the expanded dense test and from 0.74 to 0.82 on the sparse set. The sparse compound-cluster paired gain is 0.076 (95% CI 0.001–0.206), but both variants already give perfect ranking metrics and zero leave-one-compound-out threshold error and rejection false-positive rate. STCC therefore remains the default. It is a two-knob, peak-picking-free, solver-free measure with both knobs set by spectroscopy rather than by tuning.

Keywords: NMR spectroscopy · HSQC · sameness testing · spectral similarity · higher-order structure · chemical-shift perturbation

Nuclear magnetic resonance (NMR) spectroscopy is one of chemistry’s most powerful probes of matter, reading out the structure and local environment of individual nuclei non-destructively and at atomic resolution — from small organic molecules and natural products to metabolites and folded proteins. Much of that power is exercised not by solving a spectrum *de novo* but by **comparison**:

a great deal of what NMR establishes about an unknown follows from how closely the spectral fingerprint it produces matches a reference, or from how that fingerprint changes across a series. This fingerprint-comparison axis — is this the same compound as the catalogued one, is the protein still folded the same way, has a binding site been perturbed — is at root a **sameness test**, and it is one of the pillars of NMR in chemistry, and a two-dimensional ^1H -X HSQC spectrum is close to an ideal fingerprint for it: each crosspeak reports the local environment of a single ^1H -X pair, so the pattern of crosspeaks encodes molecular structure at atomic resolution.

The comparisons that matter most divide into two families — small molecules and proteins — and both turn on the same quantitative question. For small molecules, metabolites and natural products, a ^1H - ^{13}C spectrum is matched against a reference, the operation at the heart of 2D-NMR-based dereplication^[1–3] and of searches against community libraries such as the BMRB, NMRShiftDB and HMDB.^[4–6] For proteins, an amide ^1H - ^{15}N fingerprint is tracked across a ligand titration to read out binding through chemical-shift perturbation in fragment- and structure-based drug discovery,^[7–10] and — increasingly — is compared between samples to confirm that a protein’s **higher-order structure (HOS)** is preserved. Because a protein’s biological activity, stability and immunogenicity are governed by its folded secondary, tertiary and quaternary structure and not by its amino-acid sequence alone, establishing that this higher-order structure is unchanged — between a biosimilar and its originator, across a manufacturing or formulation change, or on storage — has become a central requirement of biopharmaceutical characterization and of regulatory comparability.^[11,12] Two-dimensional HSQC is a premier atomic-resolution probe of precisely that structure: two fingerprints coincide only when the two samples share the same fold, and 2D ^{13}C - and ^{15}N -HSQC fingerprinting is now an established, high-precision method for the HOS comparability of therapeutic proteins and monoclonal antibodies.^[13–15] Resolution is why HSQC, rather than a lower-resolution technique, carries this burden: methods such as circular dichroism, infrared spectroscopy and size-exclusion chromatography report structure largely as a global average,^[11] whereas an HSQC fingerprint resolves the local environment of essentially every ^1H -X site at once and therefore registers the subtle, localized change in fold or dynamics that an averaged readout can miss^[13,14] — the very differences a sameness test must catch.

Across all of these settings the same primitive is required, and it is deceptively basic: a single number that is high when two HSQC spectra come from the same species and low when they do not — the scoring core of a sameness test, exactly the measurement that multivariate and chemometric HOS workflows already carry out,^[16] and the primitive the present work makes robust. Getting that one number right is deceptively hard, because a sameness test has to resolve differences that are both small and consequential from the ordinary run-to-run variability of the measurement: a genuine change in fold, a low-level degradant or a single perturbed residue must register above analytical noise, solvent- and temperature-dependent peak drift, and the slow drift of the reference material itself, while an innocuous change must not.^[11,12] Posed precisely, “are these two the same?” is a question of *equivalence* rather than of difference — the province of the two-one-sided-tests procedure^[17] and of interval-based equivalence testing^[18] — and it is the same logic that underpins the regulatory *comparability* of a biologic before and after a manufacturing change^[19] and the analytical-similarity assessment of a biosimilar against its reference product.^[20] Thresholding that number is what turns a similarity score into a same/different verdict, so the useful measure of a sameness test is not the raw score but how cleanly it separates same-species from different-species pairs.

Three families of measure are in current use, and they fail in two opposite ways. The bin method of Bodis et al. integrates each spectrum over an $n \times n$ grid at a range of resolutions and reports

a weighted Jaccard (Ružička) index,^[21,22] in the lineage of the binning and normalization schemes long used in NMR metabonomics.^[23,24] Because its bin edges are hard, a peak straddling an edge splits its intensity across neighbouring bins; a small titration drift therefore erodes the score in a discontinuous jump. The measure is **shift-brittle**. The quad-tree method of Castillo et al. encodes a spectrum as a recursive centre-of-mass tree and compares trees node-by-node with a shift-tolerant node score,^[25] and the peak method of Pierens et al. picks peaks and matches each to its nearest neighbour in the other spectrum.^[26] Both build in genuine shift tolerance — but in a crowded window every peak has *some* near neighbour and every spectrum a similar mass-centre structure, so both saturate near unity and retain almost no contrast between related and unrelated spectra. They are **shift-blind**. A measure fit for real HSQC comparison must tolerate the small physical drifts a titration or a solvent change induces, yet still penalize intensity that is genuinely relocated.

We resolve this tension with the Shift-Tolerant Correlation Coefficient (STCC), a measure that keeps the physically meaningful ingredients of all three predecessors — area-weighted integration and unit normalization (Bodis), a physical shift tolerance (Castillo), and a lineshape picture of the spectrum (Pierens) — and adds a mean-centred correlation as the discriminating step. Given two spectra compared inside a common window Ω (processed Bruker data from any standard pipeline^[27–31] or peak lists rendered to a grid), STCC proceeds in three steps (Figure 1a).

Render. Negative intensities are clipped, each point is weighted by its local integration area, and *both* spectra are histogrammed onto a **single shared grid**, so they are directly comparable irrespective of their native digital resolution. The mass is normalized to unit sum; the final score is invariant to this scale.

Blur. Each image is convolved with a separable Gaussian whose width per axis is the physical linewidth combined with the expected drift, $\sigma = \sqrt{\ell^2 + d^2}$ — an assumed, effective lineshape (a Gaussian standing in for the true peak shape and deliberately widened by the expected drift), not a fitted one, so “lineshape” here names this modelled kernel rather than a measured profile. This replaces the bin method’s hard edges with a continuous response. For an isolated peak on a large grid, two copies displaced by (Δ_2, Δ_1) correlate as $\exp(-\Delta_2^2/4\sigma_2^2 - \Delta_1^2/4\sigma_1^2)$ — smooth and monotone in the drift (Figure 1b), so a physical shift costs little while a random relocation ($\Delta \gg \sigma$) costs almost everything. The blur width, not a bin count, is the tolerance knob, and it is set by two spectroscopic quantities: the linewidth (the floor) and the expected shift perturbation (the tolerance).

Score. Each image is mean-centred and the similarity is the zero-lag normalized cross-correlation — the Pearson coefficient — of the two centred images, clamped to $[0, 1]$. This is the mean-centred (Pearson) counterpart of the cosine / contrast-angle similarity long used in mass-spectral library search^[32,33] — equivalently, the zero-mean normalized cross-correlation (ZNCC) standard in image registration and template matching^[34] — applied to blurred 2D-HSQC images; the un-centred cosine we report as an ablation is exactly that contrast angle, and mean-centring is what separates STCC from it. Mean-centring is what turns overlap into discrimination: a cell where one spectrum has a peak (a value above its mean) and the other is empty (a value below its mean) contributes a **negative** product and lowers the score, so intensity that is not co-located is actively penalized rather than merely ignored, as it is by the one-way nearest-neighbour distance. No shift search is performed. Aligning the images — as in NMR peak-alignment tools such as *icoshift*^[35] — would let a genuinely different spectrum slide into registration and re-saturate, reintroducing exactly the failure mode of the tree and nearest-neighbour methods. Scoring at zero lag keeps the position information that carries the discrimination.

Experimental local-contrast extension. We also tested a post hoc candidate that changes only the feature used for scoring, not the STCC render or its physical shift tolerance. For the existing Gaussian-blurred image G , it forms

$$H = \sqrt{\max(G, 0)}, \quad F = H - \text{GaussianBlur}(H, 3\sigma),$$

then takes the clipped zero-lag cosine of F_x and F_y . The square root reduces domination by a few intense peaks, an intensity-weighting rationale used in mass-spectral library search but not direct evidence of NMR performance;^[39] the fixed 3σ background subtraction removes broad baseline and common low-frequency density. The factor three is fixed by design rather than exposed as another tuning parameter. The candidate adds no dependency, alignment, peak picking or model and preserves the same fixed-window comparison. Because it was developed after inspecting these benchmarks, we label it **experimental** and keep STCC as the default unless held-out threshold error or rejection false-positive rate improves without degrading the other regime.

The measure has the properties one expects of a similarity: self-similarity is exactly one, the score is exactly symmetric, and it is monotone in the drift of an isolated peak (all verified in the test suite; Supporting Information). The implementation is pure NumPy,^[36] reads processed Bruker data directly, and requires neither peak picking nor an external solver. On a representative dense pair, STCC takes 12.8 ms and the experimental local-contrast extension takes 17.2 ms (1.34 $\times$; Apple Silicon, single core). A 46-test suite (all passing) checks self-similarity, symmetry, boundedness, related-beats-unrelated ordering, monotonicity in drift, intensity-scale and render-grid invariance, explicit invalid-input errors, and the existing regressions; figures use Matplotlib.^[37] A full derivation with proofs is given in the Supporting Information.

We benchmarked STCC against the three reference methods on two deliberately **opposite** regimes, so that any measure claiming general utility must succeed on both (Figure 2; Table 1). We summarize discrimination by the **separation** (mean same-class score — mean different-class score) and the **margin** (worst same-class — best different-class score, which is what a single classification threshold — the operating point of a sameness test — actually sees); both are reported on each method’s own $[0, 1]$ scale. Every method self-scores exactly 1.00. Within each benchmark, every method and candidate is evaluated in the same fixed ppm window; no method-specific automatic overlap or alignment is used.

Dense protein ^1H – ^{15}N . The reference is the apo spectrum of the phosphatase PRL3, compared with six spectra of the same protein along a ligand titration (which should score high) and with a **different** protein (which should score low), inside a ^1H 6.5–10 $\times$ ^{15}N 105–130 ppm window. STCC scores the same protein 0.94 (range 0.89–0.97) and the different protein 0.18, a margin of 0.71 — three times the bin method’s margin of 0.24, the strongest baseline. Mean-centring earns that gap: replacing it with an un-centred cosine overlap on the *same* images drops the separation from 0.75 to 0.59. This gain is not a tuned artefact; it holds across the entire physical blur range and even survives coarsening the grid to the bin method’s own resolution (Supporting Information).

A second decoy protein and an independent titration. Because that margin is measured against a single decoy protein, we tested whether STCC’s advantage survives a harder, self-contained design: an independent PRL3 apo $\rightarrow$ olsalazine titration (23 points, from a 500 μM sample) scored against **two** distinct decoy folds — OAA and the kinase-domain EphB3 — inside the same window (Table 2; Supporting Information). As ligand is added the chemical-shift perturbation grows and STCC decays **smoothly** from 0.98 to a plateau near 0.82 — its graded shift tolerance acting on a real

titration — while both decoys sit far below every titration point (OAA 0.19, EphB3 0.41). STCC keeps a separation of 0.57 and margin of 0.41: both decoy proteins clear *every* same-protein score by a comfortable gap, whereas the tree and nearest-neighbour margins are a knife-edge (0.12 and 0.02, a single noisy titration point from crossing). The un-centred cosine trails (0.46 / 0.35), so mean-centring still earns the gap against a genuine negative distribution. This test is harder than the single-decoy benchmark on both axes — a longer titration with larger shifts and a second, tougher decoy — so STCC’s dense-regime margin is not an artefact of one negative spectrum. The experimental Local-Contrast extension reduces the two decoy scores to 0.02 and 0.14 and increases separation and margin to 0.67 and 0.52, respectively (Table 2).

Sparse small-molecule ^1H - ^{13}C . To probe the opposite regime — and to replace the single negative spectrum above with a large negative set — we used public HSQC peak lists from the simpleNMR example set^[38] for five compounds each recorded twice (one nominal sixth pair was excluded on inspection as a byte-identical duplicate; Supporting Information), and asked each method to score the 5 same-compound pairs above the 40 different-compound pairs, inside a ^1H 0–10 $\times$ ^{13}C 0–165 ppm window. The peak lists are rasterized as sticks, after which each method applies its own shift-tolerance processing; this removes the earlier double blur. Among the primary methods STCC again leads (mean same 0.746, mean different 0.0016, separation 0.745, margin 0.374). The bin method is far stronger here than on the dense fingerprint (0.29): with few, well-separated peaks its hard bins rarely straddle a peak. The tree and nearest-neighbour methods, however, **leave the two classes barely separable in this regime too** (margins 0.07 and 0.04, despite separations of 0.41 and 0.10): over a wide window a different molecule still supplies a near neighbour for every peak. Their shift tolerance is real — per pair, it keeps two shifted recordings of the same compound high (0.83–1.00, where STCC scores 0.39–0.42) — but it tolerates *everything*, so it never becomes discrimination. STCC recovers that same tolerance controllably, through its blur width, without surrendering the between-compound contrast (Supporting Information). The experimental Local-Contrast extension lifts the rotenone pair from 0.388 to 0.727 while keeping the different-compound mean at 0.0041; overall it reaches separation 0.820 and margin 0.434. Its square-root transform and local background subtraction therefore improve the descriptive sparse separation without changing peak positions or aligning spectra.

Statistical strength and held-out operation. Compound-level resampling keeps all pairs involving the same molecule together. On the current stick-input sparse benchmark, the cluster 95% interval for separation is 0.508–0.982 for STCC and 0.623–0.985 for Local Contrast. A paired compound-cluster bootstrap gives Local Contrast – STCC $\Delta = 0.0757$ ([0.0014, 0.2057], $P(\Delta \leq 0) = 0.0067$), supporting a descriptive gain on these five compounds. Ranking and threshold operation provide the promotion check: both variants have AUROC = AUPRC = top-1 = MRR = 1.00, and both have zero leave-one-compound-out classification error, false-negative rate and rejection false-positive rate. Thus the candidate does not improve the held-out decision boundary even though its separation is larger. The expanded dense result is used as regression evidence, not as an independent held-out estimate because its same-class points are one autocorrelated titration. Default promotion is reserved for lower held-out threshold error or rejection FPR without a loss in the other regime; that condition is not met here.

Across both regimes, default STCC beats the saturating tree and nearest-neighbour methods while retaining graded shift tolerance, and it matches or exceeds the bin method. The post hoc Local-Contrast extension improves descriptive separation and margin on the expanded dense and sparse benchmarks, but it does not improve held-out threshold error or rejection FPR and is therefore not promoted. STCC’s two knobs, the per-axis blur widths, are both physical: the linewidth sets the

floor and the expected shift perturbation sets how much drift is tolerated. Two caveats bound the claim honestly. First, STCC measures position-and-intensity coincidence, not molecular identity, so a close structural analogue — a near-identical dereplication candidate, or a biosimilar that differs from its originator only in a subtle structural detail — will score intermediate; this is the correct behaviour for a *similarity*, but it is also why the subtle end of higher-order-structure comparability is a task STCC scores rather than one it yet decides. Second, the three reference methods are our own representative reimplementations compared inside one common window, and the ^1H - ^{13}C spectra are rendered from peak lists; accordingly the robust **ranking**, not the exact numerical values, is the result, and this is not a verdict on the original methods as their authors deployed them with per-spectrum windows.

In summary, a single physically-motivated construction — render both spectra onto a shared grid, blur by the NMR linewidth, and take the mean-centred zero-lag correlation — reconciles shift tolerance with discrimination, the two requirements that the incumbent binning, tree and nearest-neighbour measures satisfy only one at a time. Both of STCC’s knobs are spectroscopic quantities rather than fitting parameters, its self-similarity is exact (as for every method tested), and it needs no peak picking, no alignment search and no external solver. A square-root local-contrast feature is a promising experimental extension, but the default remains the simpler validated STCC until it improves held-out operating error rather than descriptive separation alone. We are candid about scope: this is a **methodological** contribution — a robust, reproducible scoring primitive for sameness testing, not a new chemical discovery — and although its most decisive number, the dense-protein margin, was first measured against a single decoy, we now show it holds against two decoy folds on an independent titration (Table 2), alongside the fully public sparse benchmark and retrieval test. And the register is deliberately that of a scoring primitive, not a finished assay: STCC supplies the similarity score a sameness test thresholds, while fixing the equivalence bound and characterizing the false-same and false-different rates it implies — the inferential half of a formal comparability test^[17,19] — is a task the present benchmarks scope rather than settle.

Made robust, that primitive feeds straight back into the comparisons that motivated it. For small molecules it is the scoring step for library and dereplication searches and for the quality control of repeated measurements; for proteins it is the read-out for automated titration tracking and chemical-shift-perturbation mapping, and the scoring step that higher-order-structure comparability workflows require.^[15,16] On that last application we claim only what the present data establish: the dense benchmark demonstrates two of the properties an HOS read-out needs — clean separation of distinct folds (both decoys below every same-protein titration point) and a smooth, near-monotone response to a genuinely perturbed fingerprint (STCC decaying 0.98→0.82 along the titration rather than saturating) — but not the harder task at the centre of biosimilar comparability, telling a close variant from its originator, which falls in the intermediate-score regime noted above and is not demonstrated here. Characterizing where STCC places true originator/biosimilar or point-variant pairs is therefore the natural next test on the protein side, as a prospective, at-scale dereplication study against a community library is on the small-molecule side; for both, STCC now supplies the tolerant yet discriminating scoring primitive.

More broadly, deciding whether two spectral fingerprints are the same is one of the central operations by which NMR turns a spectrum into an identification or a structural verdict, and a measure that is at once tolerant of the small shifts real chemistry induces and unforgiving of genuine relocation makes that operation more trustworthy wherever it is used. Insofar as NMR remains one of chemistry’s primary windows onto matter — on molecules large and small, in solution, at atomic resolution — sharpening the way we compare its fingerprints sharpens the readout itself. The full

derivation, all three benchmark protocols, every numerical table with confidence intervals, and the 46-test suite are provided as Supporting Information; the software is freely available under the MIT license.

Figures

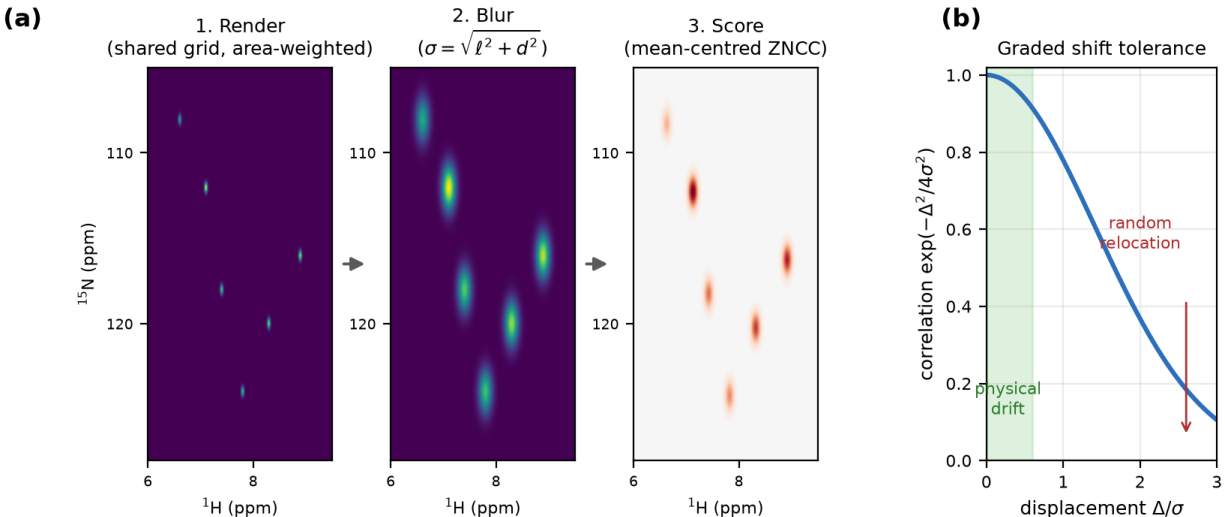


Figure 1. The Shift-Tolerant Correlation Coefficient. (a) The three-step construction: area-weighted rendering of both spectra onto one shared grid, a separable Gaussian blur of physical width $\sigma = \sqrt{\ell^2 + d^2}$ per axis, and the mean-centred zero-lag normalized cross-correlation of the two images (red/blue = positive/negative contribution to the score). (b) Graded shift tolerance: two copies of an isolated peak displaced by Δ correlate as $\exp(-\Delta^2/4\sigma^2)$, smooth and monotone in the drift, so a physical titration drift (shaded) costs little while a random relocation costs almost everything.

Table 1. Separation and worst-case margin for the primary methods on both regimes, computed from unrounded scores. Higher is better; every method self-scores 1.00.

Method	$^1\text{H}-^{15}\text{N}$ sep	margin	$^1\text{H}-^{13}\text{C}$ sep	margin
Bin (Bodis, 2009)	0.29	0.24	0.67	0.21
Bin + 45° rotation	0.25	0.20	0.67	0.19
Quad-tree (Castillo, 2013)	0.03	-0.01	0.41	0.07
Nearest neighbour (Pierens, 2012)	0.04	0.03	0.10	0.04
Cosine, un-centred (STCC ablation)	0.59	—	0.74	0.37
STCC (this work)	0.75	0.71	0.74	0.37

The un-centred cosine is STCC without mean-centring, on the same rendered/blurred images; it is an ablation, not an independent method; its dense-protein margin is not tabulated here (the code computes a margin for every method, but the stored dense ablation entry carries the separation only). The dense separations and margins in this table are `bench.py` output, regenerable once the dense spectra are deposited (Data Availability Statement). The $^1\text{H}-^{13}\text{C}$ benchmark uses 5 same-compound pairs and 40 different-compound pairs; a nominal sixth pair (olivetol) was excluded as

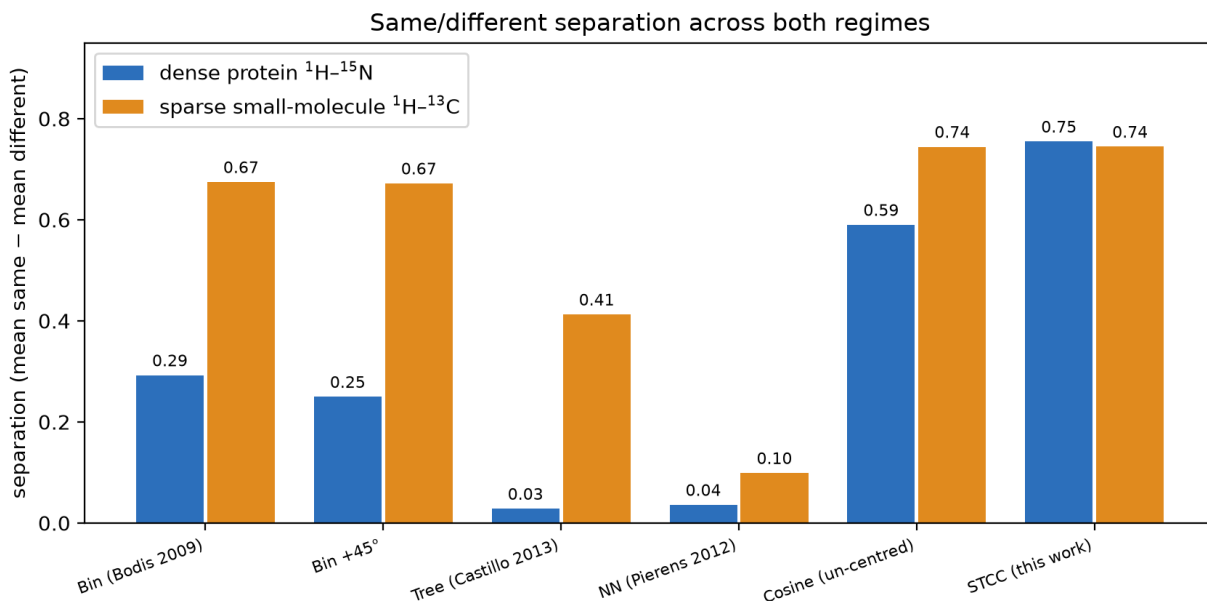


Figure 2. Same/different separation (mean same-class – mean different-class similarity) for the primary methods on the two benchmarks: dense protein $^1\text{H}-^{15}\text{N}$ (left of each pair) and sparse small-molecule $^1\text{H}-^{13}\text{C}$ (right). STCC leads the primary comparison in both regimes; the tree and nearest-neighbour methods saturate in both. The un-centred cosine (an ablation of STCC without mean-centring) matches STCC closely on the sparse regime (0.74 vs 0.74) but falls from 0.75 to 0.59 on the dense fingerprint, isolating mean-centring as the step that rescues discrimination when peaks crowd. The experimental Local-Contrast extension is reported in Table 2 and the Supporting Information rather than mixed into this original dense comparison.

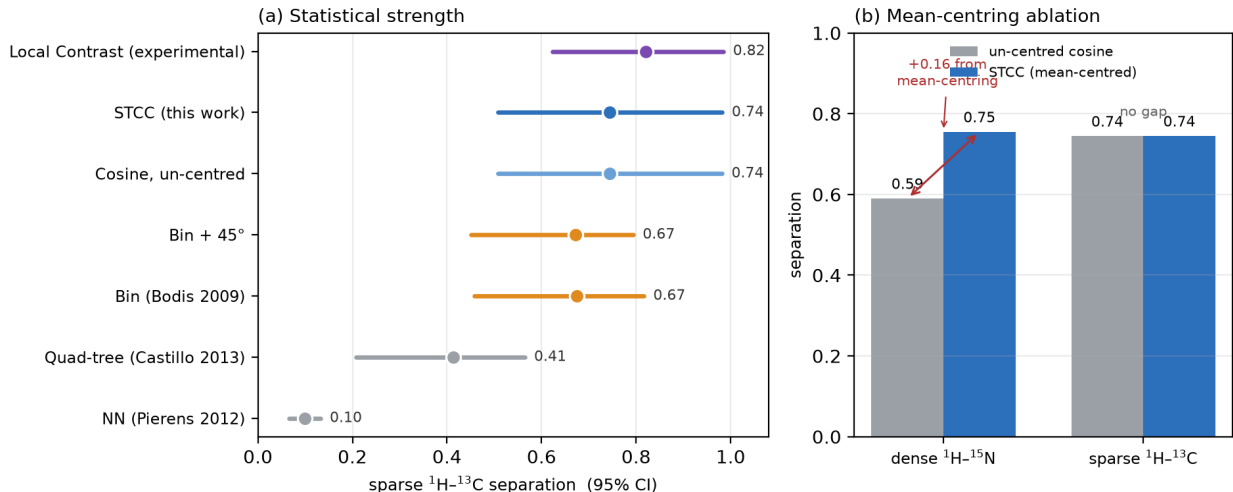


Figure 3. Statistical strength and the mean-centring ablation. **(a)** Sparse ^1H - ^{13}C separation with compound-cluster bootstrap 95% CIs (10^4 resamples of the five compounds), including experimental Local Contrast (purple). The paired Local Contrast – STCC gain is $\Delta = 0.076$ ($[0.001, 0.206]$), while both variants have identical perfect ranking and zero held-out threshold error (Table S9). **(b)** Replacing mean-centred STCC with the un-centred cosine costs 0.16 of separation on the dense protein fingerprint ($0.75 \rightarrow 0.59$) but only 0.0002 on the sparse stick set ($0.7447 \rightarrow 0.7445$), showing where global mean-centring matters.

a byte-identical duplicate. On the current stick-input sparse benchmark, compound-cluster 95% CIs are 0.508–0.982 for STCC and 0.623–0.985 for experimental Local Contrast. Their paired separation difference is $\Delta = 0.0757$ ($[0.0014, 0.2057]$, $P(\Delta \leq 0) = 0.0067$). Both nevertheless have AUROC/AUPRC/top-1/MRR = 1.00 and zero leave-one-compound-out error and rejection FPR. The candidate therefore improves descriptive separation without improving the held-out operating point.

Table 2. Expanded dense ^1H - ^{15}N benchmark addressing the single-decoy limitation: a PRL3 apo-olsalazine titration (23 same-protein points, exps 4–48) scored against **two** decoy proteins (OAA, EphB3) in the ^1H 6.5–10 $\times$ ^{15}N 105–130 ppm window. Separation = mean same – mean decoy; margin = worst same – worst decoy. Higher is better; every method self-scores 1.00. Computed by `python3 bench_nhsqc.py` (harness and derived scores `results/nhsqc_dense.json` are in the repository; raw spectra per the Data Availability Statement).

Method	mean same	min same	OAA	EphB3
Local Contrast (experimental)	0.75	0.66	0.02	0.14
STCC (default)	0.87	0.82	0.19	0.41
Cosine, un-centred (STCC ablation)	0.90	0.85	0.36	0.51
Bin (Bodis, 2009)	0.82	0.75	0.46	0.50
Bin + 45° rotation	0.86	0.81	0.52	0.56
Quad-tree (Castillo, 2013)	0.95	0.91	0.79	0.75
Nearest neighbour (Pierens, 2012)	0.99	0.98	0.96	0.96

Method	separation	margin
Local Contrast (experimental)	0.67	0.52
STCC (default)	0.57	0.41
Cosine, un-centred (STCC ablation)	0.46	0.35
Bin (Bodis, 2009)	0.34	0.25
Bin + 45° rotation	0.32	0.25
Quad-tree (Castillo, 2013)	0.18	0.12
Nearest neighbour (Pierens, 2012)	0.03	0.02

Both decoy proteins score below every one of the 23 titration points for STCC (worst same 0.82 > worst decoy 0.41) and Local Contrast (0.66 > 0.14), so each margin is positive across two distinct folds. The candidate suppresses both decoys more strongly, raising separation from 0.57 to 0.67 and margin from 0.41 to 0.52. It is nevertheless reported as an experimental, post hoc extension: the held-out sparse operating metrics below do not improve over default STCC. The STCC numbers are lower than the single-decoy benchmark (Table 1) because this test is harder on both axes: the olsalazine titration reaches larger shifts (min same 0.82 vs 0.89) and EphB3 is a tougher decoy than OAA (0.41 vs 0.19).

Supporting Information

The Supporting Information (full mathematical derivation with proofs, both benchmark protocols, all per-comparison and per-pair numerical tables, robustness sweeps, parameter reference and test-suite description) is available and referenced throughout the text.

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The author thanks the developers of the simpleNMR project for the public HSQC peak-list example set used in the sparse-regime benchmark.

Conflict of Interest

The author declares no conflict of interest.

Data Availability Statement

The software (STCC, experimental Local Contrast, the three reference methods, the un-centred-cosine ablation, all benchmark harnesses, the tabulated scores and the 46-test suite) is freely available under the MIT license at https://github.com/deepnmr/spectra_similarity_tool. The **sparse ^1H - ^{13}C benchmark and the dereplication/retrieval experiment are fully reproducible from public data**: the harnesses download the simpleNMR peak lists (<https://github.com/EricHughesABC/simpleNMR>) automatically and contain no hard-coded paths. The dense ^1H - ^{15}N benchmark uses PRL3 and OAA Bruker spectra available from the author on reasonable request (the harness reads their location from `--prl3/--oaa` or `$PRL3_DIR/$OAA_DIR`); a public deposition of the processed spectra is in preparation so that the dense benchmark also becomes independently reproducible. The expanded dense benchmark of Table 2 (PRL3-olsalazine titration and the OAA and EphB3 decoy spectra) is computed by `bench_nhsqc.py` from the same processed Bruker data and is included in that deposition.

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Table of Contents Entry

Tolerant yet discriminating. Testing whether two 2D-HSQC spectra are the same demands tolerance of small chemical-shift drifts *and* penalization of genuinely relocated intensity — requirements the incumbent binning, tree and nearest-neighbour measures satisfy only one at a time. The Shift-Tolerant Correlation Coefficient renders both spectra onto one grid, blurs them by the physical NMR linewidth, and takes the mean-centred zero-lag correlation. An experimental local-contrast extension further improves descriptive separation across dense protein and sparse small-molecule examples, while the validated STCC remains the default.

Suggested TOC graphic: the separation bar chart (Figure 2) with the three-step STCC schematic inset.